Helpdesk Ticket Lite: Project Write-up

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Demo video link:

https://drive.google.com/file/d/1bp6lSpGERD9u9HWNajdRKbv4NPTy2-bg/view?usp=sharing

GitHub link:

https://github.com/mary-jofina-jose/helpdesk-ticket-lite

1. Business problem

Employees need a simple way to report basic IT issues such as a laptop that will not boot or Wi-Fi that is down. The helpdesk needs to route each issue to a technician, keep an eye on deadlines, and close the ticket with the user's feedback. The MVP covers raising and editing tickets, assignment to a technician, an SLA due date based on priority, a feedback request when a ticket is resolved, and a dashboard showing ticket volume and resolution performance.

2. Solution overview

Everything was built inside one Dataverse solution called Helpdesk Ticket Lite (publisher prefix hd) in a Power Apps developer environment.

Layer	Tool	What it does
Data	Dataverse	Ticket, Technician and Comment tables, choice columns, lookups and an autonumber ticket ID
App	Power Apps (canvas)	Browse, Details, Edit and Assistant screens with search and filters
Automation	Power Automate	Alert and SLA due date on new tickets, technician email on assignment, feedback approval on resolution
Analytics	Power BI	KPI cards, volume by category, priority and status, SLA breached percentage
Stretch	Rule-based assistant	Answers common questions and opens a pre-filled ticket form

3. Data model

Ticket: Title, Ticket Number (autonumber, HD-0001), Description, Category (Network, Hardware, Software, Access & Login, Other), Priority (Low, Medium, High, Critical; default Medium), Status (New,

Assigned, In Progress, Resolved, Closed; default New), Requester Email, Assigned To (lookup to Technician), Due Date, Resolved On, CSAT Rating (1 to 5) and CSAT Comment.

Technician: Name, Email, Specialty (Network, Hardware, Software, Other) and Active.

Comment: Comment Text, Ticket (lookup to Ticket), Commented By and Internal Note.

Relationships

- One Technician has many Tickets.
- One Ticket has many Comments.

Comments are a separate table because a ticket collects several notes from different people over time. Each comment is its own row with an author and an automatic timestamp, so the history stays ordered and nothing is overwritten.

4. Canvas app

The app has four screens.

- **Browse:** a gallery of tickets with number, title, status and assigned technician. A search box and status and priority filters narrow the list. A coloured badge shows priority, a chip shows status, and a red OVERDUE tag appears when a ticket is past its due date and not yet Resolved or Closed.
- **Details:** a display form for one ticket with a status banner that reads Awaiting assignment, Assigned, Resolved or Closed. A comments panel lists notes and lets a technician add new ones. Once a ticket is Closed, the banner shows the star rating and comment, and editing is switched off.
- **Edit:** one form for creating and updating tickets. Due Date, Resolved On and the CSAT fields are read-only, because the flows set them.
- **Assistant:** a chat-style screen that answers common questions such as how to reset the VPN from a keyword list. When it has no answer, or the user taps Raise a ticket, it opens the Edit form with the question pre-filled in the title.

5. Automation (Power Automate)

Flow 1: Ticket Created, Alert and SLA. It triggers when a ticket row is added. It reads the ticket, maps the priority to SLA hours (Critical 4, High 8, Medium 24, Low 48), writes the due date back to the ticket, and emails the helpdesk coordinator with the ticket details, the SLA target and the due time in IST.

Flow 2: Ticket Assigned, Notify Technician. It triggers when the Assigned To column changes. It reads the technician's row and emails them the ticket details, priority and due time.

Flow 3: Ticket Resolved, CSAT. It triggers when the Status column changes. If the status is Resolved and Resolved On is empty, the flow:

1. Sets Resolved On to the current time.
2. Emails the requester that the ticket is resolved.
3. Starts an approval with custom responses 1 to 5 and a comment box, and waits for the answer.

4. Saves the rating and comment on the ticket.
5. Sets the status to Closed when the rating is 3 or more, or back to In Progress and clears Resolved On when the rating is 1 or 2.

Each trigger is narrowed to one event or one column, and Flow 3 checks that Resolved On is empty, so a flow's own update does not start another run.

6. Analytics (Power BI)

The report connects to Dataverse in Import mode. Choice columns arrive as numbers, so calculated columns turn them into labels. Created Actual uses the backdated created time where available, and otherwise works it out from the due date minus the SLA hours.

- KPI cards: total tickets, average resolution time in hours, SLA breached percentage and average CSAT.
- Bar chart of tickets by category and column chart of tickets by priority.
- A funnel and a donut chart of tickets by status.
- Visuals cross-filter, so clicking a category updates every card and chart.

SLA breached % is the number of resolved tickets whose Resolved On is later than their Due Date, divided by all resolved tickets. **Average resolution time** is the average gap between created time and Resolved On, for resolved tickets only.

For the charts, 21 backdated sample tickets were imported from Excel with the flows switched off, so no emails were sent and the imported due dates were not overwritten.

7. Stretch goal: Helpdesk Assistant

The brief suggests a Copilot Studio agent. The tenant has no Copilot Studio licence, so the stretch goal was built as a rule-based assistant inside the app. A small keyword list answers questions such as VPN reset, Wi-Fi, printer and password problems. If no keyword matches, or the user taps Raise a ticket, the standard ticket form opens with the question pre-filled. The ticket then follows the same pipeline as any other: alert, SLA, assignment and feedback.

8. Acceptance criteria

Criterion	Result
New ticket created and saved	Met
Alert on ticket creation	Partly met. Sent by email, because the tenant has no Teams licence
Assignment routes to a technician	Met. The technician is emailed on assignment
Simple SLA timer	Met. The due date is set from the priority, and the app shows an OVERDUE tag
Resolution triggers feedback	Met. A 1 to 5 approval is sent when the status

Criterion	Result
	becomes Resolved
Feedback captured end to end	Met. Rating and comment are saved on the ticket, and the ticket closes or reopens
Search and filter by status and priority	Met
Analytics: volume by category and priority, average resolution time	Met
Stretch: Copilot Studio agent	Not met as specified. Delivered as a rule-based assistant

9. Limitations and assumptions

- **Teams:** the free developer tenant has no Teams licence, so alerts are sent by email. The flow can be switched to a Teams post by replacing one step.
- **Email sender:** emails are sent through the Gmail connector, so they come from a personal address and may land in spam.
- **Approvals:** approvals go to tenant users, so the approval was assigned to the work account. In a real company the requester would be a directory user.
- **Single test user:** testing used one account acting as requester, coordinator, technician and approver.
- **Assistant:** it is rule-based and only knows the keywords in its list. A Copilot Studio agent would replace it given a licence.
- **Delegation warning:** filtering on choice columns shows a delegation warning in the app. This is acceptable for a small MVP dataset.
- **Sample data:** the imported sample tickets are unassigned, because the technician lookup was left out of the import.
- **Power BI refresh:** the report uses imported data and needs a refresh to show new Dataverse data.

10. Next steps

1. Post alerts to a Teams Helpdesk channel.
2. Assign tickets automatically by technician specialty.
3. Notify the requester when the assigned technician changes the status.
4. Replace the rule-based assistant with a Copilot Studio agent that creates tickets from chat.
5. Add scheduled refresh for the Power BI report.

11. Screenshots

Tables

Ticket table (custom columns)

Ticket columns and data

Update forms and views | Edit

Title	Assigned To	Category *	CSAT Commen	+19 more
Computer Power On issue	Mary Jofina Jose	Hardware		
Excel macro error		Software		
Keyboard keys not working		Hardware		
Laptop charge issue		Hardware		
Laptop connecting issue	Mary Jofina Jose	Network		
Laptop not booting		Hardware		
Laptop not booting	Mary Jofina Jose	Hardware		
Laptop not booting		Hardware	Fixed fast, thank y	
Login Denied	Mary Jofina Jose	Access and lo...	Quick fix, thanks!	

Technician table

Technician columns and data

Update forms and views | Edit

Name	Technician *	Email *	+18 more
Anil Kumar	bcbb6bf6-18d8-f111-84ea-7ced8...	abc@gmail.com	
Mary Jofina Jose	bdbb6bf6-18d8-f111-84ea-7ced8...	maryjofinajose@gmail.com	
Meera Nair	fba5ae99-01d9-f111-84ea-7ced8...	abcmeera@gmail.com	
Rahul Das	4c066004-19d8-f111-84ea-7ced8...	abcrahul@gmail.com	
Enter text		*Enter email*	

Comment table

Comment columns and data

Update forms and views | Edit

Created On	Comment *	Comment Text	+18 more
11/9/2026 1:18 PM	59274be3-11dc-f111-84ea-7ced8...	power adapter replaced ..check n...	
11/9/2026 8:42 AM	b0e07456-ebdb-f111-84ea-7ced8...	Power adptor is replaced now try i...	
11/5/2026 12:55 PM	2a42e304-ead8-f111-84ea-7ced8...	Replaced the power adapter, testi...	
11/5/2026 1:19 PM	26af6e73-edd8-f111-84ea-7ced8...	Replaced the power adapter, testi...	
11/5/2026 1:25 PM	9f4a854a-eed8-f111-84ea-7ced8...	Replaced the power adapter, testi...	
11/4/2026 8:30 PM	7ea66c79-60d8-f111-84ea-7ced8...	Restarted the Router	
		Enter text	

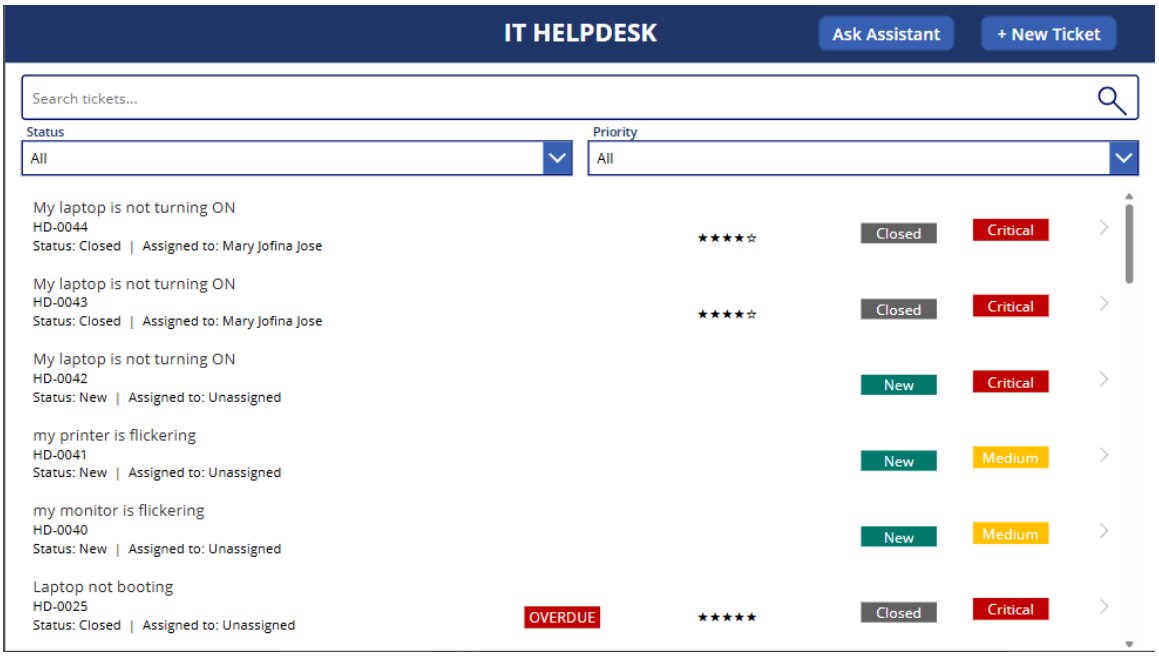
Relationships (Ticket table)

Helpdesk Ticket Lite > Tables > Ticket > **Relationships**

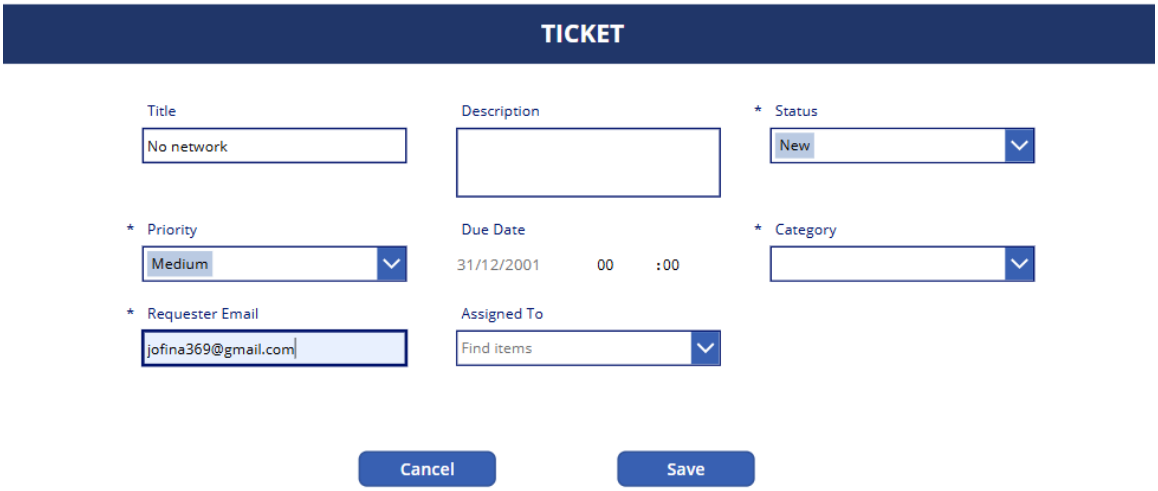
Display name	Name	Related ta...	Relationshi...	Managed	Customized	Customizable
Assigned To	hd_ticket_Assign...	Technician	Many-to-one	No	Yes	Yes
Ticket	hd_comment_Tic...	Comment	One-to-many	No	Yes	Yes

Canvas app

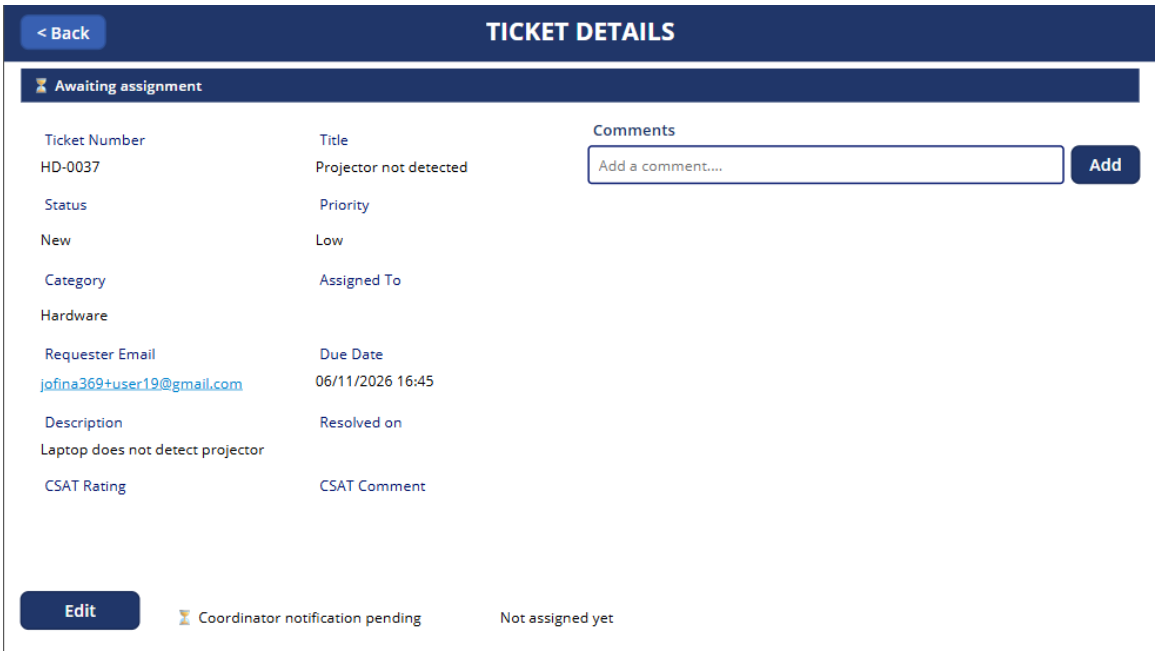
Browse screen



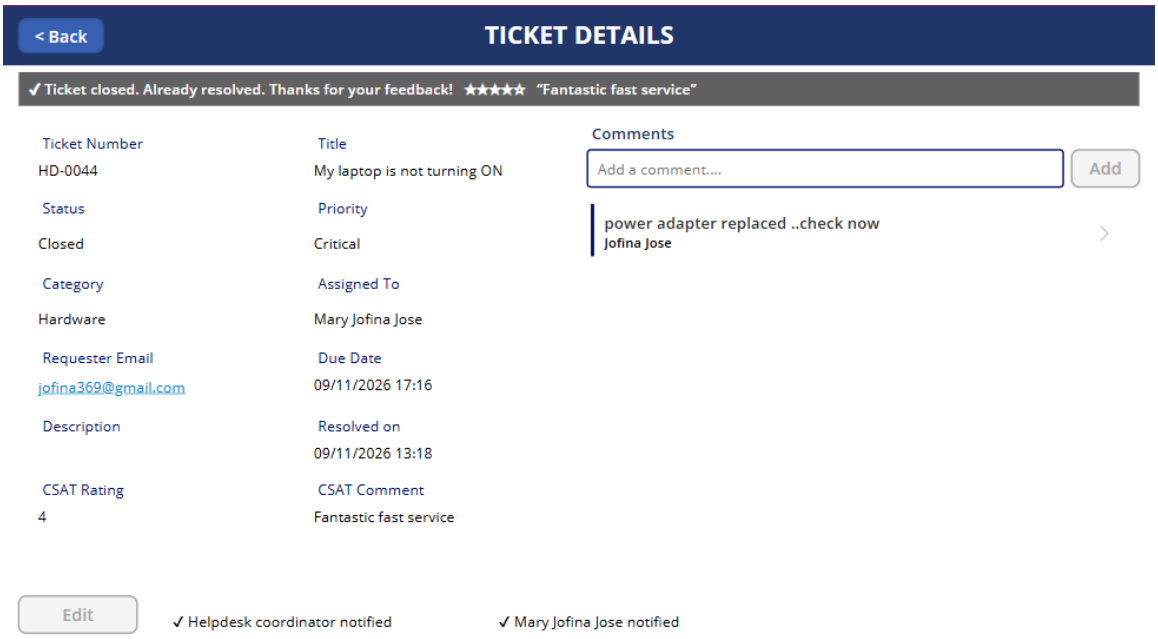
Edit screen



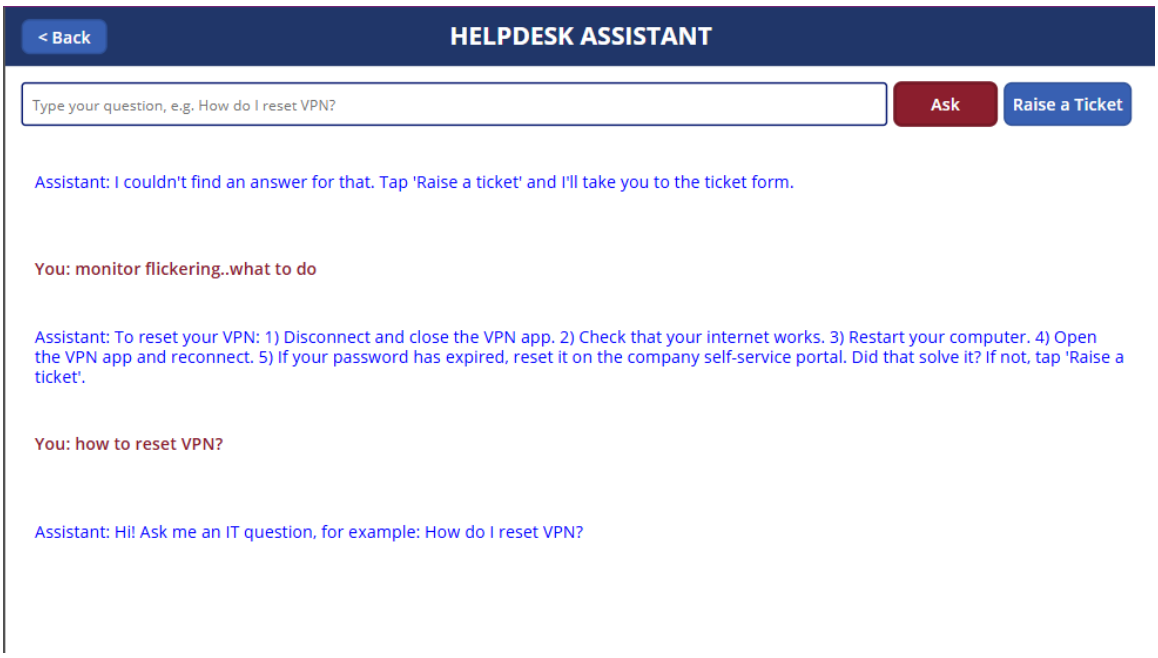
Details screen (open ticket)



Details screen (closed ticket with rating)



Assistant screen



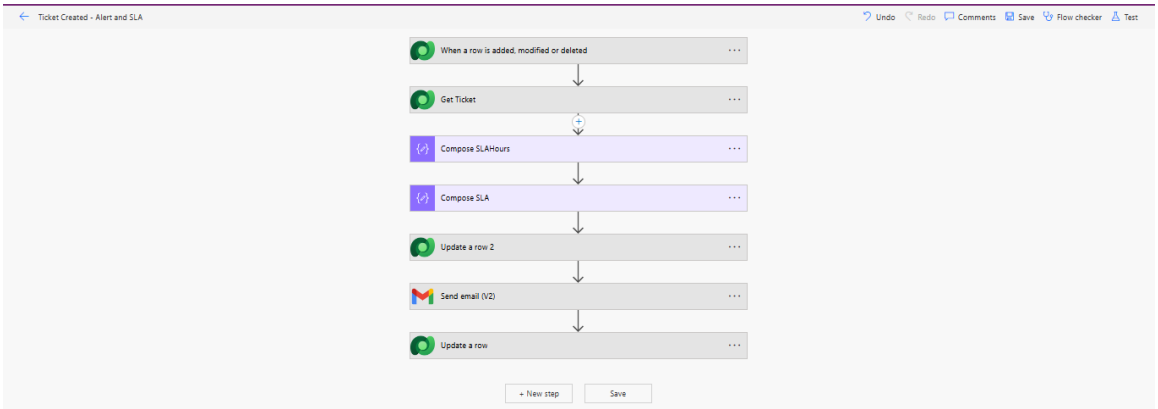
Cloud flows

Flow list

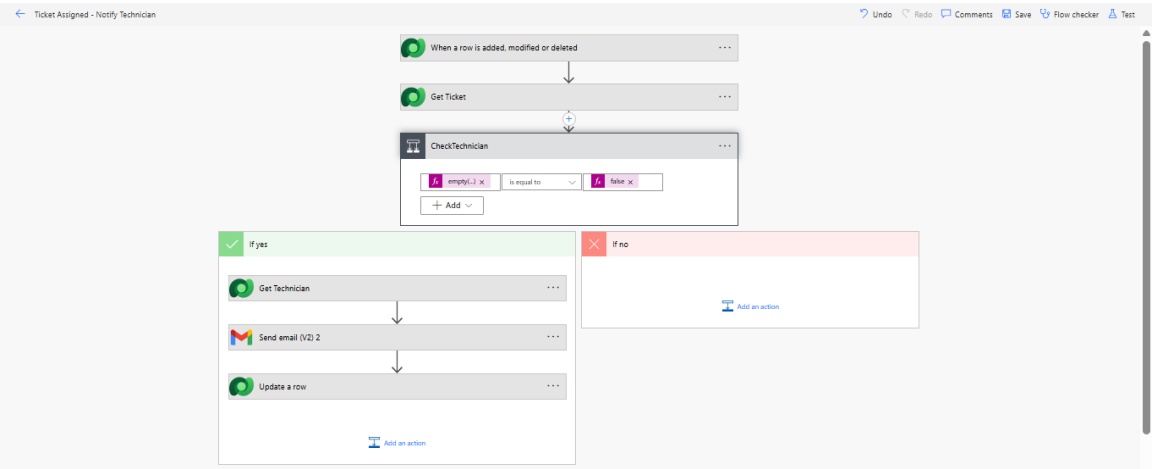
Helpdesk Ticket Lite > **Cloud flows**

Display name	Name	Type	Managed	Customized	Last Modif...	Owner
Ticket Assigned - Noti...	Ticket Assigned -...	Cloud Flow	No	Yes	3 days ago	# Jofina Jose
Ticket Created - Alert ...	Ticket Created - ...	Cloud Flow	No	Yes	3 days ago	# Jofina Jose
Ticket Resolved - CSAT	Ticket Resolved -...	Cloud Flow	No	Yes	3 days ago	# Jofina Jose

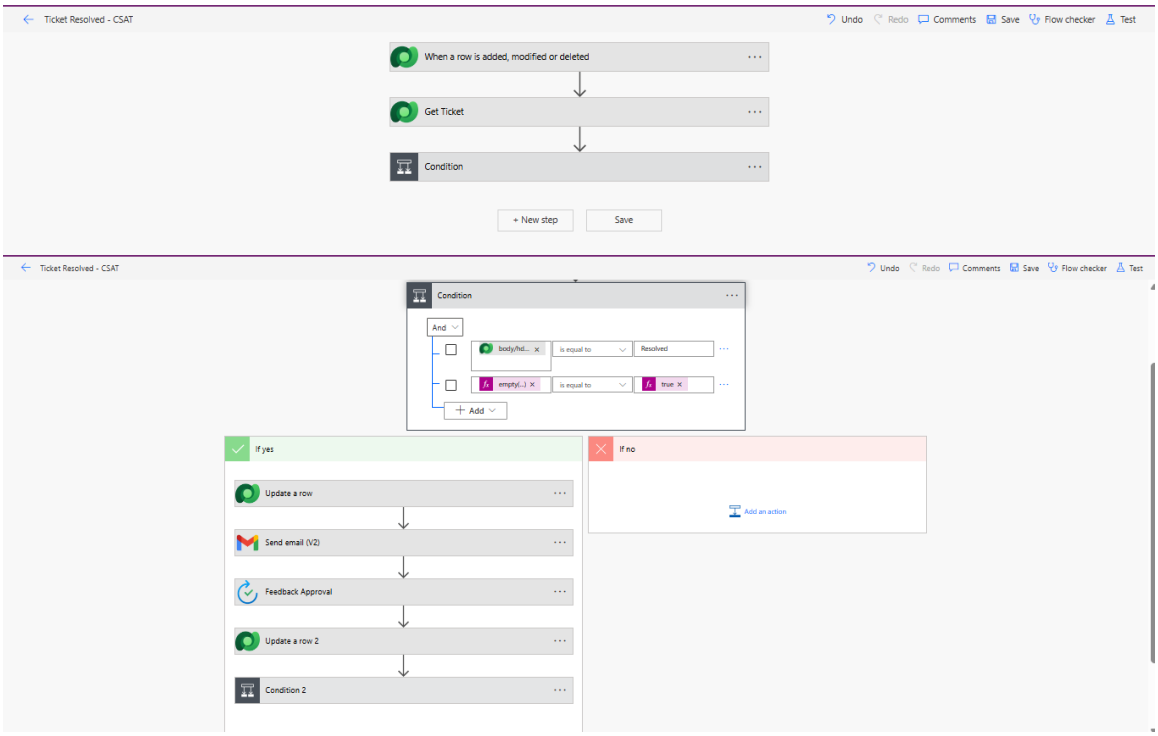
Flow 1: Ticket Created, Alert and SLA



Flow 2: Ticket Assigned, Notify Technician

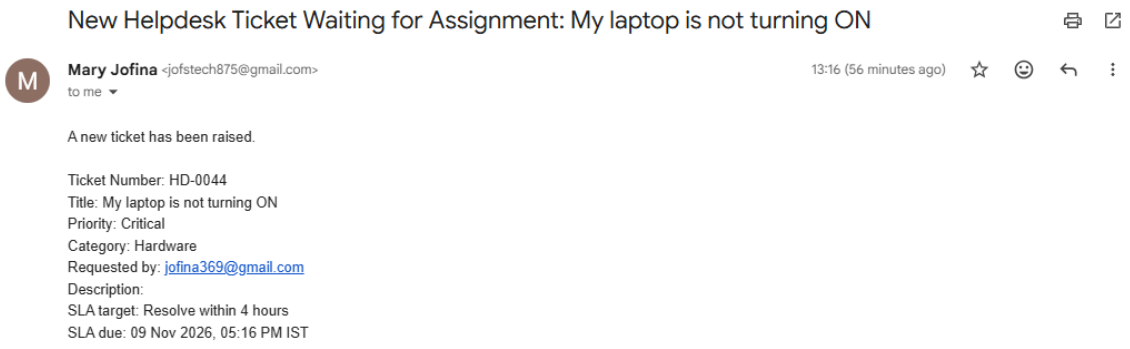


Flow 3: Ticket Resolved, CSAT

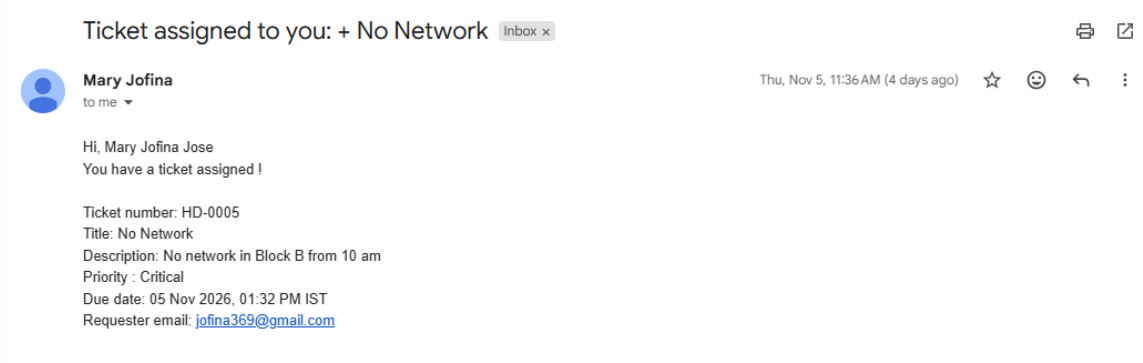


Emails and approval

New ticket email



Technician assignment email



Feedback approval request



Power BI report

Ticket analytics dashboard

